CALIBRATION CATALOGUE

Dimension

Basic Instruments

Every manufacturedpart startswith
a measurement.
From a vernier caliper on the shop floor
to a micrometer in the quality lab,
these instruments decide whether
a component fits, functions and
passes inspection.

REVA PHOENIX
CALIBRATION & CONSULTANCY

DIMENSION BASIC INSTRUMENTS

TRUSTED MEASUREMENT PARTNER

Calibration & Consultancy

Keepingevery instrumentaccurate and every audit confident.

01 Traceable accuracy

02 Audit-ready compliance

03 Expert guidance

25 YEARS OF TRUSTED EXPERTISE

REVA PHOENIX
CALIBRATION & CONSULTANCY

DIMENSION BASIC INSTRUMENTS

WHY DIMENSIONAL CALIBRATION MATTERS

01 — **Part Fit &Function:**Even a fewmicronsof errorcan turn a goodcomponent into scraporcause assembly failures downstream.

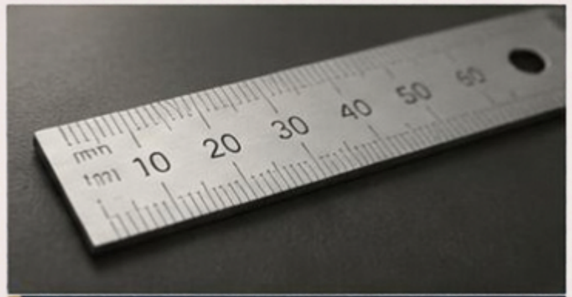
02 — **Right Quality Decisions:** Inspection results are only as reliable as the instrument behind them. Calibration keeps decisions correct.

03 — **Lower Cost of Quality:** Fewer rejections, less rework and reduced warranty and recall risk protect your margins.

04 — **Audit Confidence:** Traceable calibration supports ISO 9001, IATF 16949, AS9100 and ISO/IEC 17025 requirements.

Calibration is not a cost. It is insurance for every measurement you make.

Instruments at a Glance



Steel Rule

Quick linear checks and layout reference. refer:

Vernier Caliper

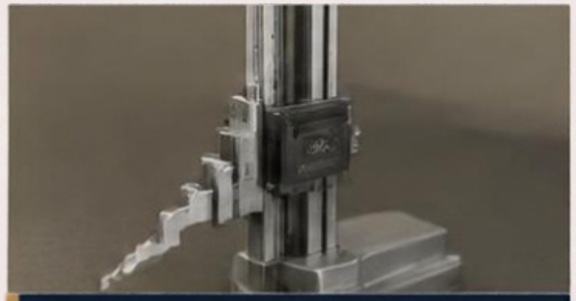
Outside, inside, depth and step measurements.

Micrometer

High-resolution outside diameter and thickness.

Dial Gauge

Comparison, run-out and fitness checks.



Height Gauge

Height measurement and layout marking.

Feeler Gauge

Gap and clearance measurement.

Plug (Go / No-Go) Gauge

Fast pass / fail check of hole size.

Gauge Blocks

Reference standard of length.

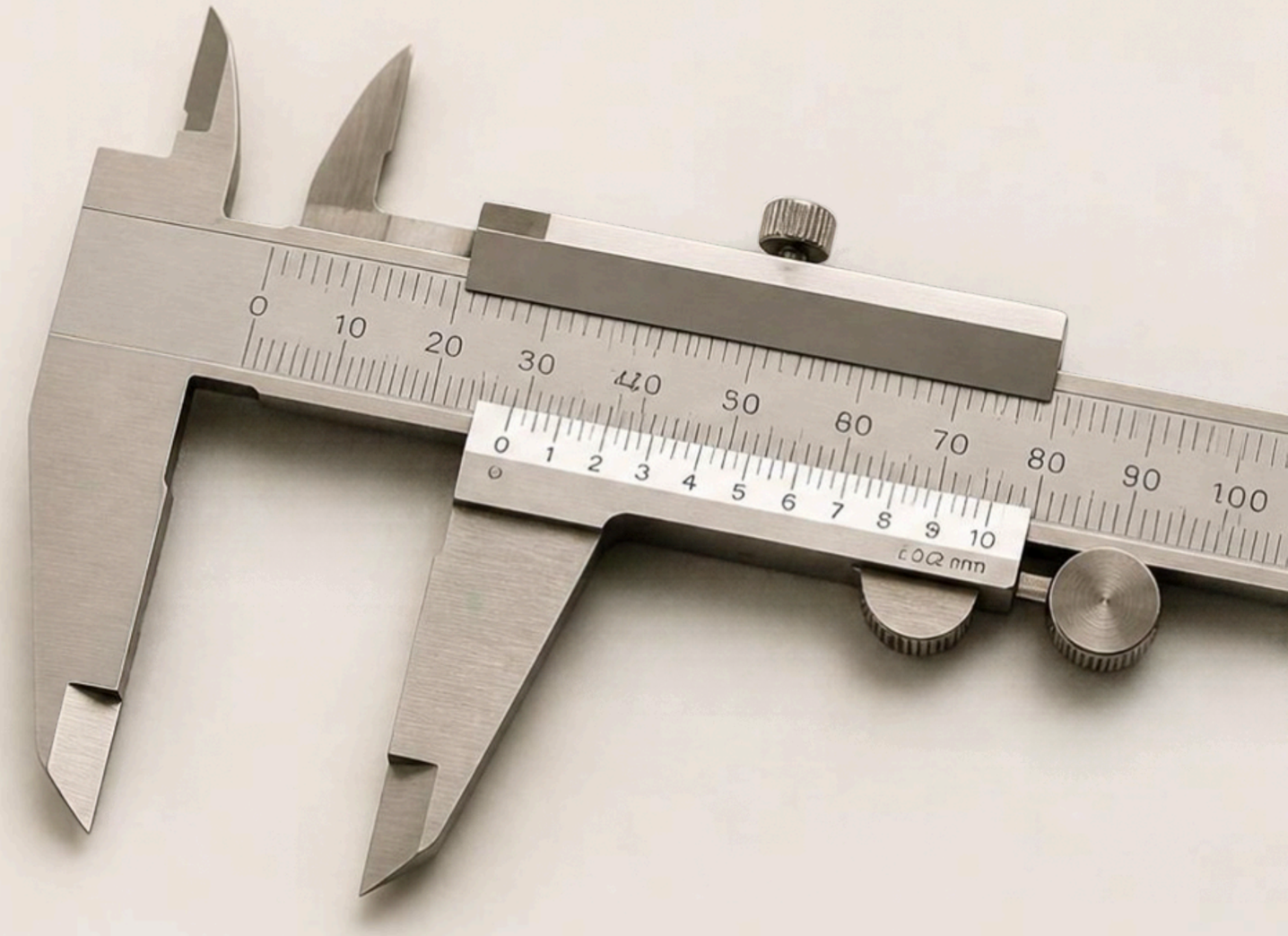
VERNIER CALIPER

HOW IT WORKS: Themain scalegiveswhole millimetres while the sliding vernier scalerevealsthefraction. The jaws capture the feature and the readingistaken where the two scales align.

0–300 mm (Common range); 0.02 mm (Typical least count); OD · ID · Depth (Measures).

What we calibrate:

Zero error with jaws closed
Accuracy of outside, inside and depth readings at multiple points
Jaw parallelism, flatness and wear
Scale and vernier condition, slider smoothness
Repeatability of readings



REFERENCE standards: Grade 0 / 1 gauge blocks (ISO 3650), setting rings and pin gauges, with calibration at the 20 °C reference temperature.

Micrometer

HOW IT WORKS: A precision screw turnsthe spindle by a known pitch.The sleeveshows whole units and the thimble shows thefinereading, so tiny movements become large, readable scale divisions.

0–25 mm (Typical frame size);
0.01 mm (Typical least count);
Ratchet (Constant force).

What we calibrate: Zero error and setting standard check;
Flatness and parallelism of anvil and spindle faces;
Accuracy at several points across the range;
Spindle runout and thimble backlash;
Ratchet / friction-thimble force consistency.

REFERENCE STANDARDS:

Gauge blocks (ISO 3650), optical flats and optical parallels for face flatness and parallelism, in a temperature-controlled laboratory.

DIAL GAUGE & HEIGHT GAUGE

DIAL GAUGE

Converts a small plunger movement into a magnified needle reading. Ideal for comparison, run-out and flatness checks.

Accuracy across full range

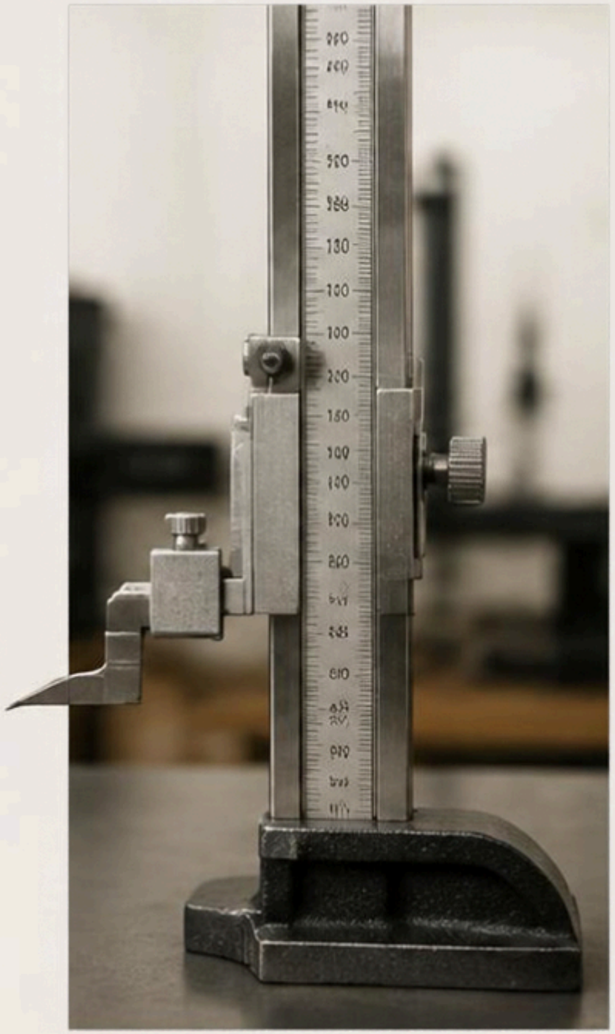
Repeatability and hysteresis

Plunger movement and return

Resolution typically 0.01 mm

HEIGHT GAUGE

A precision column with a sliding scriber or probe for height measurement and layout marking from a reference surface.



Scale accuracy along the column

Squareness of column to base

Scriber / probe condition

Digital and vernier types

OUR CALIBRATION PROCESS

1. **Receive & Identify:** Instrument is logged, identified and its condition recorded.
2. **Inspect & Clean:** Visual and functional checks. Cleaned and stabilised at 20 °C.
3. **Measure:** Compared against traceable reference standards at defined points.
4. **Evaluate:** Errors and measurement uncertainty are calculated and recorded.
5. **Certify & Label:** Certificate issued, calibration label and due date applied.

WHAT YOU RECEIVE:

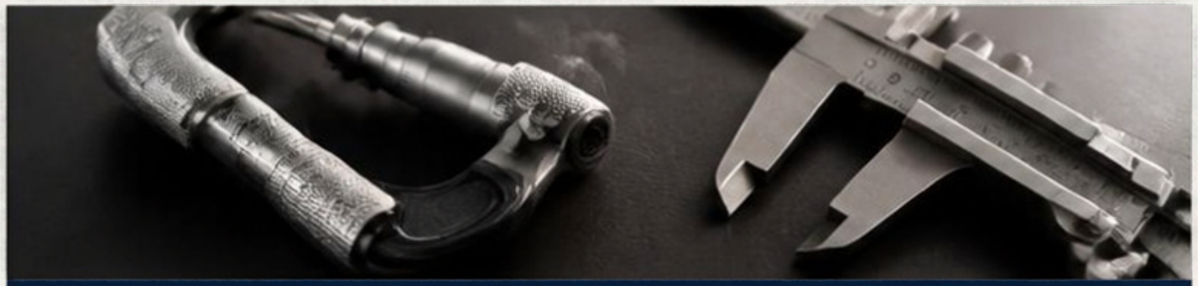
Calibration certificate with measured values and uncertainty

Traceability to national / international standards

Calibration label showing the next due date

Clear guidance on adjustment, repair or replacement

When to Calibrate & How to Care



Send it for calibration when...

- The zero reading does not return to zero
- Jaws, anvils or faces show wear or damage
- The instrument was dropped or overloaded
- Readings are inconsistent or drifting
- The calibration due date has been reached
- A critical inspection or audit is coming

Care that protects accuracy

- Wipe clean after every use
- Store in its case, away from dust and oil
- Never force jaws or spindle; use the ratchet
- Keep away from heat and direct sunlight
- Handle gently. Do not use as a tool
- Record every calibration and due date

Typical calibration interval: 6 to 12 months. Your exact interval depends on usage, criticality and instrument history, and we help you set it.

WHY REVA PHOENIX

Measure with **confidence.**

25+ years of engineering and quality experience

Training, consulting, implementation, engineering and auditing under one roof

Serving automotive, oil & gas, aerospace and business excellence

Global standards. Real results.

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